

Swing: 2D and 3D Animation in Virtual Reality

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ABSTRACT

Swing is a narrative film in virtual reality space combining 2D, 3D and 2.5D animation techniques. The story unfolds in overlapping acts that depict the internal and external struggles of a frustrated girl who is attempting and failing to swing on a playground swing. As she struggles with the swing, she imagines the demise of the children swinging successfully around her, and anger builds until she ends up taking it out on herself. When she finally hits rock bottom, she lets go, leans back and swings. A new perspective is found.

Keywords: VR, narrative, 2D animation, Oculus Quill animation, 3D animation, Unreal, Maya, virtual reality.

1 THE INFERNO

After the opening sequence of *Swing*, viewers are immersed in the animated mind of a girl on a playground who has become frustrated with herself for not being able to swing like the other children.



Figure 1: Opening Act of *Swing*, before entering the Inferno

The 3D-animated characters (representing the “real” world) become increasingly overshadowed by a coliseum of arches housing 2D-animated looping souls in a pre-teen take on Dante’s Inferno.

Proximity to the loops as they rise from the ground, the souls’ stories can be heard via sound effects and short voice clips. These stories are fictionalized versions of anonymously-collected accounts of jealousy as described by a diverse population in an online survey. The viewer can follow the narrative or explore the soul stories in the arches.

As the other children succeed, the viewer descends into the young girl’s personal hell. When the girl begins attacking herself, she is momentarily launched into the air in a false and unsustainable flight. Her sobering return to the ground casts a literal light on her situation, and in looking towards that light she leans back, prompting the swing to swing.

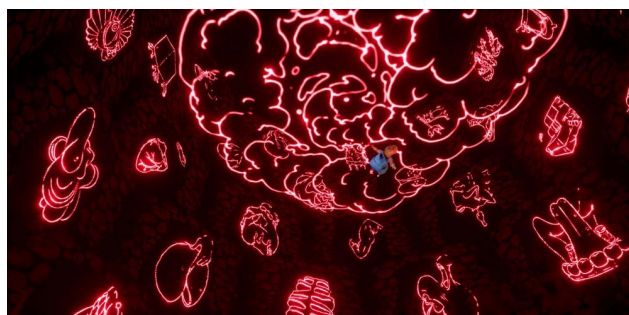


Figure 2: The Inferno takes over, a tunnel of 2D animated loops

2 RATIONALE

While “Swing” is superficially about children on a playground, its overarching theme of struggle leading to discovery springs from the experiences of two real women overcoming adversity in seemingly small but impactful situations. This theme was partnered with virtual reality’s singular ability to immerse viewers in an experience, and the recognition that VR stories based on emotion are less common than those focused more on action. This opportunity for empathetic viewing greatly altered the original format of this story, but the resulting iteration, which enabled the inclusion of over fifty diverse stories, is that much richer.

3 CONCLUSION

Virtual Reality offers new and exciting avenues for immersive storytelling not only in regards to environment but also human experience. In *Swing*, we are able to place the viewer inside someone else’s head, creating an opportunity for representation through voice and image. We are also exploring how the viewer engages with 360-degrees of inter-weaving animation in the hopes that all iterations convey the thematic heart of the film.

REFERENCES

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